

Homework-03

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Set Up

```
# load in packages
library(tidyverse)
library(janitor)
library(readxl)
library(here)
library(ggeffects)
```

```
# store kelp data as an object called kelp
kelp <- read_csv(here("data", "temp-kelp.csv"))
# store egg data as an object called egg_data
egg_data <- read_csv(here("data", "personal-data-project - Sheet1.csv"))
```

Problems

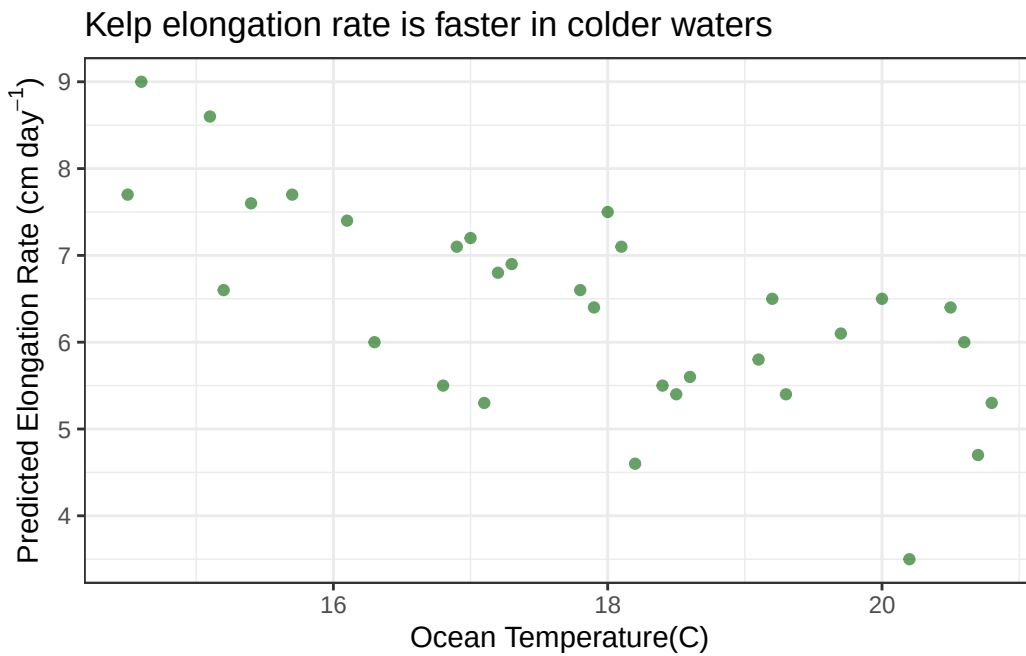
Problem 1: Giant Kelp Fronds

a. An appropriate test

To determine the strength of the relationship between temperature and elongation rate, you could do a Pearson's correlation and a Spearman rank correlation. A Pearson's correlation is parametric and is best for understanding the strength of the correlation between variables when there is a linear relationship between continuous variables. A Spearman's rank correlation is non-parametric and useful for understanding the strength of the correlation between variables that show a monotonic relationship between variables.

b. Create a visualization

```
# base layer: ggplot
ggplot(data = kelp, # use kelp data frame
       aes(x = temp_c, # x-axis: temperature (C)
           y = kelp_elong)) + # y-axis: kelp elongation rate (cm/day)
# 1st layer: show raw data points
geom_point(color = "darkgreen", # points = dark green
           alpha = 0.6) + # points = slightly transparent
# label and y axes
labs(x = "Ocean Temperature(C)",
     y = expression(paste("Predicted Elongation Rate (cm ", day-1,")")),
# add title
     title = "Kelp elongation rate is faster in colder waters") +
# change theme
theme_bw() +
# remove legend
theme(legend.position = "none")
```



c. Check assumptions and run test

Create Linear Model

```
# create new object called kelp model to store linear model
# function: lm
# formula: dependent ~ independent, data = data frame
kelp_model <- lm(kelp_elong ~ temp_c, data = kelp)
```

```
# show summary of linear model
summary(kelp_model)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8643	-0.5017	0.1790	0.5659	1.2177

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.08645	1.49219	9.440	1.72e-10 ***
temp_c	-0.43179	0.08321	-5.189	1.37e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

```
flextable::as_flextable(kelp_model)
```

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	14.086	1.492	9.440	0.0000 ***
temp_c	-0.432	0.083	-5.189	0.0000 ***

Estimate	Standard Error	t value	Pr(> t)
Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05			

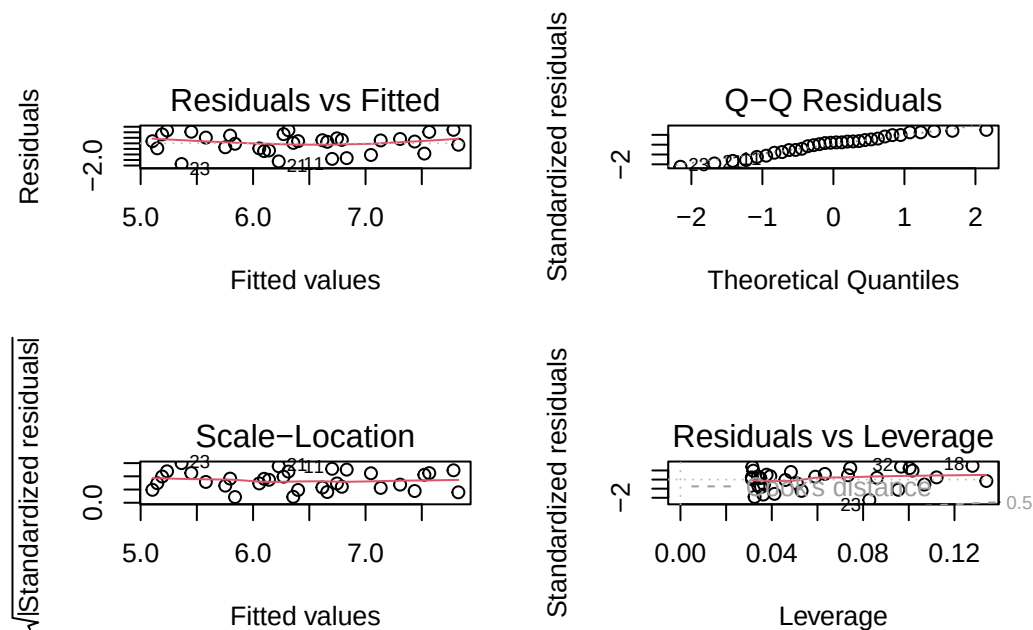
Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 30 and 1 DF, p-value: 0.0000

Check Assumptions

```
# create 2x2 grid to show diagnostic plot
par(mfrow = c(2,2))
# show diagnostic plots
plot(kelp_model)
```



The QQ residuals plot shows the data points following the line. This suggests the distribution of the residuals is normal. The residuals vs. fitted and scale locations plots don't seem to show a pattern. The data seems to be randomly and evenly distributed which indicates that the residuals are homoscedastic. The residuals vs leverage plot does not show points outside of the dashed lines which indicates that there are no outliers influencing the model.

Run Test

```
kelp_cor <- cor.test(kelp$temp_c, kelp$kelp_elong, method = "pearson")
show(kelp_cor)
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and kelp frond elongation, I used Pearson's correlation (r). This test is a good choice because both ocean temperature and elongation rate are continuous variables, and the diagnostic checks show that the data satisfies the assumptions of normality and linearity. I suspect a linear relationship between ocean temperature and kelp frond elongation rate

I found a strong negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's $r = -0.7$, $t()$)